

GroAurum

Admin Inventory Management v1

Warehouse stock console with append-only movement ledger

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apps/admin-web · UI shell only · Phase 3 adapters (af9e616)

This is NOT a simple stock table. Operators manage warehouse positions, reservations, and an append-only ledger - not a spreadsheet.

1. Purpose

Provide warehouse operators and administrators with a complete stock management console inside the GroAurum Admin shell. Sidebar navigation is unchanged; Inventory replaces the placeholder module.

2. Design Principles

- Professional SaaS, desktop-first, high information density
- GroAurum Design System - white cards, green branding, 8px spacing
- Warehouse operations console feel - not a spreadsheet
- Append-only movement ledger - balances derived, history never overwritten
- Typed fixtures only - no backend mutations in v1

3. Inventory Dashboard

Route: /inventory

- Total Stock Value
- Available Stock
- Reserved Stock
- Incoming Stock
- Low Stock Alerts

4. Inventory Table

Column	Meaning
SKU	Name + code - links to detail
Warehouse	Stock location / hub
Available	Free to sell
Reserved	Held for open orders
Incoming	Supplier receipts pending
Reorder Level	Threshold for low alerts
Status	Healthy / Low / Out of Stock / Incoming

5. SKU Detail

Route: /inventory/:skuld

- Overview - warehouse balances, reorder level, stock value, ledger footnote
- Movement History - append-only timeline of stock events
- Reservations - stock held for wholesale orders
- Adjustments - cycle counts, damage write-offs, clearance

6. Stock Movement Types

- Supplier Receipt
- Customer Order
- Damage
- Manual Adjustment

Movements form an append-only ledger. Historical rows retain their values; available / reserved / incoming are derived views.

7. Inventory Status

- Healthy - above reorder level
- Low - at or below reorder level
- Out of Stock - zero available
- Incoming - inbound supplier stock in flight

8. Quick Actions

- Receive Stock - focuses Movement History (mutation deferred)
- Adjust Stock - focuses Adjustments tab
- View Ledger - opens Movement History

9. Data Preparation

```
apps/admin-web/src/data/inventory-types.ts
apps/admin-web/src/data/inventory-fixtures.ts
# INVENTORY_SNAPSHOT_FIXTURE + INVENTORY_DETAIL_FIXTURES
# Replace with Supabase inventory / ledger queries
```

10. Files

- pages/inventory/InventoryListPage.tsx - KPI dashboard + table + actions
- pages/inventory/InventoryDetailPage.tsx - tabs for overview / ledger / reservations
- components/inventory/* - table, badges, timeline, reservations, adjustments, actions
- components/dashboard/KpiCards.tsx - shared KPI cards (widened for inventory)
- App.tsx - /inventory and /inventory/:skuld routes

11. Validation

Check	Result
pnpm --filter @groaurum/admin-web typecheck	Pass
pnpm --filter @groaurum/admin-web build	Pass
Sidebar / Products / Pricing	Unchanged
Backend mutations	Not implemented (by design)

12. How to Review

```
pnpm dev:admin
# Open http://localhost:5173/inventory
# AKH-LA-10 - low stock + full movement ledger
# KIS-EL-CTN - incoming status
# MIX-LEG-CTN - out of stock
```

13. Out of Scope

- Supabase receive / adjust / reserve mutations
- In-place balance edits (intentionally forbidden)
- Barcode / WMS device workflows
- Multi-warehouse transfer UI

14. Intent

Inventory Management should feel like a warehouse operations console: see stock value and risk at a glance, inspect SKU positions by hub, and trace every change through an append-only ledger - never a flat editable grid.